

INVERSE LAPLACE

- 1) $L^{-1}\left\{\frac{1}{s}\right\} = 1$
- 2) $L^{-1}\left\{\frac{1}{s-a}\right\} = e^{at}$
- 3) $L^{-1}\left\{\frac{1}{s+a}\right\} = e^{-at}$
- 4) $L^{-1}\left\{\frac{a}{s^2+a^2}\right\} = \sin at$
- 5) $L^{-1}\left\{\frac{a}{s^2-a^2}\right\} = \sinh at$
- 6) $L^{-1}\left\{\frac{s}{s^2+a^2}\right\} = \cos at$
- 7) $L^{-1}\left\{\frac{s}{s^2-a^2}\right\} = \cosh at$
- 8) $L^{-1}\left\{\frac{1}{s^n}\right\} = \frac{t^{n-1}}{(n-1)!} \text{ or } \frac{t^{n-1}}{n!}$

TYPES

- 1) CONVOLUTION THEOREM
- 2) PARTIAL FRACTION
- 3) EXAMPLES ON LOG AND INVERSE
- 4) DIVISION BY 'S'

TYPE 1: CONVOLUTION THEOREM

$$L^{-1}\{f_1(s)f_2(s)\} = \int_0^t f_1(u) f_2(t-u)du$$

STEPS:

- 1) SPLITTING
- 2) NAMING
- 3) INVERSE
- 4) FORMULA

$$1) \quad \sin A \cos B = \frac{1}{2} [\sin(A + B) + \sin(A - B)]$$

$$2) \quad \cos A \sin B = \frac{1}{2} [\sin(A + B) - \sin(A - B)]$$

$$3) \quad \cos A \cos B = \frac{1}{2} [\cos(A + B) + \cos(A - B)]$$

$$4) \quad \sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$$

$$5) \quad \sin(-x) = -\sin x$$

$$6) \quad \cos(-x) = \cos x$$

$$7) \quad \sin^2 x = \frac{1 - \cos 2x}{2} \quad \text{or} \quad \cos^2 x = \frac{1 + \cos 2x}{2}$$

Rules for hyperbolic

I) Same Formula लिखो

II) 'h' add करो

III) $\sin \times \sin$ hai तो sign change करो

for example

$$\sinh A \sinh B = \frac{1}{2} [-\cosh(A - B) + \cosh(A + B)]$$

OR

$$-\sinh A \sinh B = \frac{1}{2} [+ \cosh(A - B) - \cosh(A + B)]$$

TYPE 3: DIVISION BY 'S'.

$$L^{-1} \left\{ \frac{1}{s} f(s) \right\} = \int_0^t f(u) \, du$$

TYPE 4: EXAMPLES ON LOG AND INVERSE.

$$L^{-1} \{ f(s) \} = \frac{-1}{t} L^{-1} \{ f'(s) \}$$

$$1) \quad \frac{d}{dx} \log f(x) = \frac{1}{f(x)} \frac{d}{dx} [f(x)]$$

$$2) \quad \frac{d}{dx} \tan^{-1} f(x) = \frac{1}{1 + f(x)^2} \frac{d}{dx} [f(x)]$$

$$3) \quad \frac{d}{dx} \cot^{-1} f(x) = \frac{-1}{1 + f(x)^2} \frac{d}{dx} [f(x)]$$

$$4) \quad \tanh^{-1} x = \frac{1}{2} \log \frac{1+x}{1-x}$$